

Quantum Federated Learning With Decentralized Data

Rui Huang^{ID}, Xiaoqing Tan^{ID}, and Qingshan Xu^{ID}

Abstract—Variational quantum algorithm (VQA) accesses the centralized data to train the model, and using distributed computing can significantly improve the training overhead; however, the data is privacy sensitive. In this paper, we propose communication-efficient learning of VQA from decentralized data, which is so-called quantum federated learning (QFL). Motivated by the classical federated learning algorithm, we improve data privacy by aggregating updates from local computation to share model parameters. Here, aiming to find approximate optima in the parameter landscape, we develop an extension of the conventional VQA. Finally, we deploy on the TensorFlow Quantum processor within variational quantum tensor networks classifiers, approximate quantum optimization for the Ising model, and variational quantum eigensolver for molecular hydrogen. Our algorithm demonstrates model accuracy from decentralized data, which have higher performance on near-term processors. Importantly, QFL may inspire new investigations in the field of secure quantum machine learning.

Index Terms—Quantum algorithm, quantum computing, quantum information, quantum machine learning.

I. INTRODUCTION

TRADITIONAL machine learning is based on sample database training to obtain different tasks and scenarios. These databases are collected and stored centrally from different users, terminals, and systems. In practical application scenarios, this centralized data training approach faces some challenges. One is to compromise the privacy and security of data, and the other is to increase the communication overhead. Federated learning [1] allows multiple users, called clients, to collaborate

on training a shared global model without sharing data on the local device. Interestingly, it eliminates the need to centralized aggregate data on a single device, allowing the machine learning model to learn different data on each client. Existing studies have demonstrated a range of federated learning applications, such as horizontal federated learning [2], vertical federated learning [3], and federated transfer learning [4]. A prominent example of these algorithms is knowledge sharing without compromising user privacy [5].

Quantum machine learning (QML) uses quantum characteristics to speed up classical machine learning algorithms. One key is that QML performs fast linear algebra or quantum Fourier transforms in the state space. To some extent, QML achieves exponential or quadratic acceleration compared with classical algorithms [6], including supervised and unsupervised learning [7], [8]. In this context, these algorithms are based on quantum random access memory [9], and universal implementations on quantum hardware are of concern. These works study QML from the perspective of quantum acceleration and excavate the possible quantum advantages.

The noisy intermediate-scale quantum (NISQ) processor [10] will help explore many-body quantum physics, and quantum algorithms for quantum devices in the near term have emerged. This new field of research is called parameterized quantum circuits (PQC) [11], similar to the heuristic algorithms of classical machine learning. One key to learning a particular task is to optimize the parameters of the circuit, where the objective function is constantly approaching the target. At the same time, the variational quantum algorithm (VQA) has emerged, which is applied to many-body quantum physics and quantum chemistry. We note that the variational algorithm includes variational quantum tensor networks (VQTN) classifiers [12], quantum approximate optimization algorithm (QAOA) [13], and variational quantum eigensolver (VQE) [14].

In the field of VQA, such an approach is well suited for the NISQ era. A novel quantum tensor network algorithm is proposed, which uses kernel encoding to implement shallow quantum circuits, which has higher performance on near-term processors [12]. A quantum algorithm based on collective optimization is designed, which solves a group of related Hamiltonian effectively [15]. However, the current research mainly focuses on centralized data training to obtain different tasks and scenarios. Inspired by the above algorithm design, we set our sights on the field of secure quantum machine learning. In this paper, we focus on the quantum algorithm running on near-term devices and demonstrates the solution of discriminant

Manuscript received August 30, 2021; revised March 17, 2022; accepted April 22, 2022. Date of publication April 26, 2022; date of current version May 12, 2022. This work was supported in part by the Open Fund of Advanced Cryptography and System Security Key Laboratory of Sichuan Province under Grant SKLACSS-202106, in part by the Natural Science Foundation of Guangdong Province of China under Grant 2021A1515011440, in part by the Major Program of Guangdong Basic and Applied Research under Grant 2019B030302008, in part by the National Natural Science Foundation of China under Grant 62032009, and in part by the Outstanding Innovative Talents Cultivation Funded Programs for Doctoral Students of Jinan University under Grant 2021CXB007. (Corresponding author: Xiaoqing Tan.)

Rui Huang and Qingshan Xu are with the College of Information Science and Technology/College of Cyber Security, Jinan University, Guangzhou, Guangdong 510632, China (e-mail: huang6666@foxmail.com; 1301817455@qq.com).

Xiaoqing Tan is with the College of Information Science and Technology/College of Cyber Security, Jinan University, Guangzhou, Guangdong 510632, China, and also with the Advanced Cryptography and System Security Key Laboratory of Sichuan Province, Chengdu, Sichuan 610025, China (e-mail: ttanxq@jnu.edu.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSTQE.2022.3170150>.

Digital Object Identifier 10.1109/JSTQE.2022.3170150

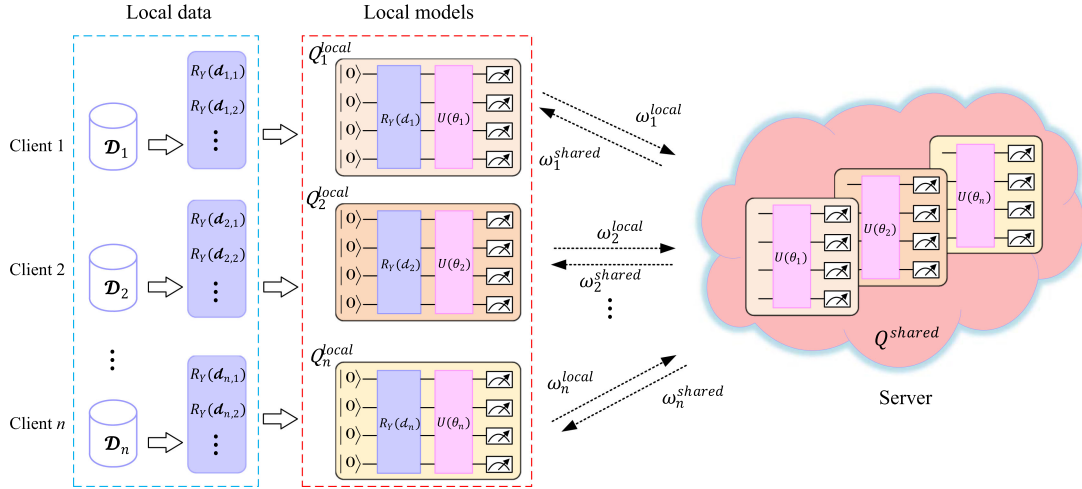


Fig. 1. Illustration of the QFL scheme. Each local classical data $\mathcal{D}_N$, $\forall N \in \{1, 2, \dots, n\}$, which is encoded into quantum data through a Y -rotation gate, and the data satisfies non-i.i.d. According to the proposed application scenario, the local models consists of the corresponding VQA, initializes the variational parameters, and predicts the model results with the input quantum data. The server then aggregates the received parameters and shares the results with all local clients.

classifier, combinatorial optimization, and molecular simulation problems. Our work has two significant contributions. Firstly, we propose a quantum federated learning (QFL) algorithm, aiming at communication-efficient learning of VQA from decentralized data. Privacy and security risks are significantly reduced by limiting possible attacks on clients with decentralized data. At the same time, training on multiparty quantum devices is regarded as an important research direction. Secondly, we deployed a QFL experiment on TensorFlow Quantum (TFQ) processor [16], which is implemented on near-term processors. The experiment includes non-independent and identically distributed (non-i.i.d) quantum data, evaluation variational circuit model, and stochastic gradient descent. Importantly, our experiment shows that the number of communication rounds for multiparty quantum devices is reduced. Our results explore new avenues for secure quantum machine learning, which is key to the privacy and security of quantum data.

The remainder of this paper is organized as follows. In Section II, we propose QFL using VQTN, QAOA, and VQE from decentralized data. Section III puts forward the experiment for QFL on the TFQ processor and presents our conclusion in Section IV.

II. QUANTUM FEDERATED LEARNING

In this section, we present the QFL, which focuses on the privacy computing architecture of decentralized data. The federated learning process consists of a server and several clients. In each training round, the local client evaluates the VQA circuit using private data and uploads model parameters to the server. In this work, these local clients consist of quantum computers or quantum simulators. The server then aggregates the received parameters and shares the results with all local clients. The whole process of the QFL scheme is shown in Fig. 1.

Each local models Q_N^{local} , $\forall N \in \{1, 2, \dots, n\}$, receives local quantum data $\mathcal{D}_N$ and predicts the result. The local prediction

of Q_N^{local} is a set of optimized solutions in client N , and the model parameters are denoted as ω_N^{local} . The server takes ω_N^{local} as input to aggregate the shared parameters, and then outputs ω_N^{shared} to optimize Q_N^{local} .

The quantum model and scenario of federated learning are as follows. Firstly, QFL is suitable for participants with different sample sizes to solve the same problem jointly. For example, quantum devices in different regions solve classification, combinatorial optimization, and quantum chemistry problems. Secondly, each participant downloads the latest model from the server. Besides, the model gradients of local data training are encrypted and uploaded to the server, which aggregates the gradients of each user to update the model parameters. Each participant updates the local model parameters and repeats the above steps to achieve convergence.

QFL training is based on real data from multiparty quantum devices, which has obvious advantages over centralized training in the data center. At the same time, relative to the number of model parameters, data is large and privacy-sensitive. Private data cannot be logged to other devices during local training. It has several key properties that differ from distributed computing. Firstly, quantum computing performance is different between clients in the real world, so the training data satisfies non-i.i.d. Secondly, some clients train more frequently than others, so the data for local training is unbalanced. In this context, we focus on non-i.i.d and unbalanced quantum data, as well as efficient communication.

A. Related Work

Li *et al.* [17] proposed a private distributed learning framework using blind quantum computing (BQC), and numerical experiments show that it explores the potential of computational tasks with privacy guarantees. The method is that Alice wants to complete a quantum task but does not have a quantum computer, so she delegates the task to Bob, who has quantum computing

power. After the private data is encrypted locally, the brickwork state is uploaded to the server for measurement. However, the brickwork state is large, and the server contains private data during training.

At the same time, Chen *et al.* [18] studied federated training on a hybrid quantum-classical machine learning model, which aims to improve data privacy. The idea is to use VGG16 to pre-train the classic model with 224×224 pixel images, and compress the information into 4 qubits as input to VQA. The quantum circuit was then repeated twice to predict the classification. However, this method uses a classical deep learning algorithm for large-scale training and does not show quantum advantages.

Our work extends these ideas from five aspects. Firstly, we introduce non-i.i.d and unbalanced quantum data used to simulate effective real-world communication. Secondly, multiple quantum processors unit (QPU) share the same set of model parameters aggregated using classical processors unit (CPU). Thirdly, quantum stochastic gradient descent (qSGD) was introduced to avoid the analysis gradient too costly. Fourthly, combined with the adaptive moment estimation (Adam) [19], the learning rate of circuit parameters is automatically adjusted to accelerate convergence. Finally, a trained QFL is successfully deployed on near-term processors, learning to find approximate optima in the parameter landscape of VQTN, QAOA, and VQE.

B. Federated Quantum Data

For classical data as input to a quantum algorithm, one key is to define an encoding method that encodes classical data into a quantum state. There are many ways to realize quantum encoding, including analog encoding and digital encoding. Here, analog encoding stores data as amplitudes of states, whereas digital encoding as qubit strings. In traditional QML, an N -qubit quantum state needs to be constructed as follows

$$|\psi\rangle = \sum_{j=1}^N c_j |j\rangle, \quad (1)$$

where the classical input data $\{c_j\}_{j=1}^N$ is normalized to satisfy $\sum_{j=1}^N |c_j|^2 = 1$. Here, the data is encoded in the quantum state of the complex amplitude, i.e., analog encoding. However, the general implementation of this encoding method on quantum hardware is of concern. Instead of N qubits encoding N classical data [20]–[22], in this work, we use the kernel encoding method to encode the classical data into a quantum state [12], which requires $\lceil (N/2) \rceil$ qubits to encode N classical data.

Federated quantum data is non-i.i.d, and clients usually have different data distributions. Some clients have less local data and lack local training samples, while others have a lot of local data. It can be used to optimize the learning model to improve the prediction accuracy of each client. However, the heterogeneity of federated data poses new challenges in the field of multi-tasking learning. Caldas *et al.* [23] proposed a modular benchmark framework for learning in federated environments, the so-called Leaf. Based on this framework, we use Leaf to process the raw MNIST data set and encode it as federated quantum data. In

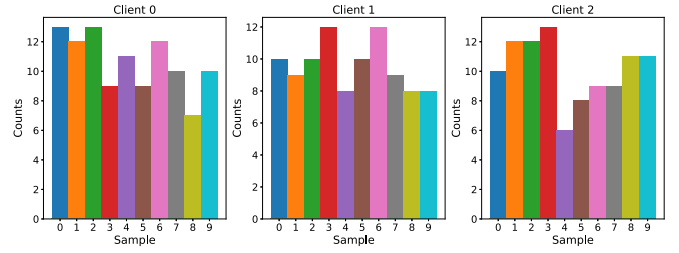


Fig. 2. Heterogeneity of federated quantum MNIST data. We visualized the number of examples of handwritten labels on each client. Firstly, the total sample size of each client is different, which reflects the local data capacity of different quantum devices. Secondly, the proportion of handwritten labels is different, showing the personalized research content of different quantum devices. In this context, the number of examples on each client can vary considerably depending on the client.

Algorithm 1: Local Model.

Input: A sequence of exponentials $\hat{U}(\theta)$, and a measurement operator $\hat{H}$.

Output: A bit string $y \in \{0, 1\}^n$.

```

1 while some condition holds do
2   1. Perform the unitary operation  $\hat{U}(\theta)$  on the initial
   state  $|\psi_0\rangle^{\otimes n}$ , namely an ansatz  $\theta \rightarrow |\Psi_\theta\rangle$ .
3   2. Measured the all terms  $\hat{H}$ , namely
    $\varepsilon(\theta) = \langle \Psi_\theta | \hat{H} | \Psi_\theta \rangle$ .
4   if Training then
5     3. Minimize the target energy and calculate the
     gradient  $\nabla J^{(t)}(\theta)$ .
6     4. Update parameters, namely
      $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta^{(t)} \nabla J^{(t)}(\theta)$ .
7   else if Testing then
8     5. return  $\varepsilon(\theta)$  and  $\theta$ .
9   end
10 end

```

order to visualize, the heterogeneity of MNIST data set [24] for non-i.i.d is shown in Fig. 2.

C. Local Models

VQA is a hybrid quantum-classical algorithm that performs multiple iterations to obtain the optimal solution according to different problem scenarios. However, conventional VQA research has focused on centralized data training for different tasks and scenarios. Here, we define the local models, including VQTN, QAOA, and VQE. The overall procedure of the local model is shown in Algorithm 1. It is divided into three stages, namely initialize ansatz, evaluation quantum circuit, and gradient calculation.

1) *Vqtn*: Let us first introduce the VQTN ansatz and then focus on the application of the classification problem. Ansatz comprises a series of unitary operators consisting of two arbitrary rotations and a XX entangled gate. Besides, the ansatz framework consists of either tree tensor network or multi-scale entanglement renormalization ansatz, depending on the specific classification problem. Here, the quantum data is initialized as

$|\psi_0\rangle^{\otimes n} = \bigotimes_{j=1}^{N_q} \mathbf{R}_y(\mathcal{D}_{N,j})|0\rangle$. The prediction then evolves into a final state $|\Psi_\theta\rangle$.

Definition 1: The local VQTN, which is calculated by Q_N^{local} for model parameter ω_N^{local} , is defined as follows

$$Q_N^{local} = \bigotimes_{j=1}^{N_q} \mathbf{XX}(q_{j+1}|q_j) \mathbf{R}_y(\omega_{N,j}^{local}) \mathbf{R}_y(\mathcal{D}_{N,j})|0\rangle, \quad (2)$$

where $\mathbf{XX}$ and $\mathbf{R}_y(\cdot)$ indicate the entangled and Y -rotation gate, respectively.

2) *Qaoa*: QAOA starts from the uniform superposition state

$$|\psi_0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \left(\sum_{\mathbf{x} \in \{0,1\}^n} |\mathbf{x}\rangle \right). \quad (3)$$

The wave function is then produced to calculate the *cost Hamiltonian* $\hat{H}_C$. Applied onto this uniform superposition state with a sequence of exponentials of the form

$$\hat{U}(\theta_m, \theta_c) = \prod_{j=1}^P e^{-i\theta_m^{(j)} \hat{H}_M} e^{-i\theta_c^{(j)} \hat{H}_C}, \quad (4)$$

where $\hat{H}_M = \sum_{j \in \mathcal{V}} \hat{X}_j$ is known as the *mixer Hamiltonian*. The final quantum state output is given by $|\Psi_\theta\rangle = \hat{U}(\theta_m, \theta_c)|\psi_0\rangle^{\otimes n}$.

Definition 2: The local QAOA, which is calculated by Q_N^{local} for model parameter ω_N^{local} , is defined as follows

$$Q_N^{local} = \prod_{j=1}^P e^{-i\omega_{N,m_j}^{local} \hat{H}_M} e^{-i\omega_{N,c_j}^{local} \hat{H}_C} |\psi_0\rangle^{\otimes n}. \quad (5)$$

For the MaxCut problem, the $\hat{H}_C$ function f is as follows

$$\hat{H}_C = f(\hat{\mathbf{Z}}) = \sum_{\{j,k\} \in \mathcal{E}} \frac{1}{2} (\hat{I} - \hat{Z}_j \hat{Z}_k), \quad (6)$$

where $\mathcal{E}$ are the edges of the graph $\mathcal{G}$.

3) *Vqe*: Here we describe the VQE ansatz by energy minimization. The Hubbard Hamiltonian as follows

$$\hat{H} = -T \sum_{\substack{(i,j), \\ \sigma \in \{\uparrow, \downarrow\}}} (\hat{a}_{i,\sigma}^\dagger \hat{a}_{j,\sigma} + \hat{a}_{j,\sigma}^\dagger \hat{a}_{i,\sigma}) + U \sum_i \hat{a}_{i,\uparrow}^\dagger \hat{a}_{i,\uparrow} \hat{a}_{i,\downarrow}^\dagger \hat{a}_{i,\downarrow}, \quad (7)$$

where $\hat{a}_{i,\sigma}$ and $\hat{a}_{i,\sigma}^\dagger$ are annihilation and creation operators respectively. T and U indicate a hopping and a many-body interaction term, respectively. The VQA ansatz is a Trotterization approximation based on time evolution, as follows

$$\hat{U}(\theta_t, \theta_u) = \prod_{j=1}^P e^{-i\theta_t^{(j)} \hat{T}} e^{-i\theta_u^{(j)} \hat{U}}, \quad (8)$$

where P is the Trotter steps.

Definition 3: The local VQE, which is calculated by Q_N^{local} for model parameter ω_N^{local} , is defined as follows

$$Q_N^{local} = \prod_{j=1}^P e^{-i\omega_{N,t_j}^{local} \hat{T}} e^{-i\omega_{N,u_j}^{local} \hat{U}} |\psi_0\rangle^{\otimes n}. \quad (9)$$

where the initial state $|\psi_0\rangle^{\otimes n}$ is a Hartree-Fock state for the quantum chemistry.

4) *Quantum Stochastic Gradient Descent*: Now, we introduce gradient optimization for local models. The gradient of hybrid quantum-classical algorithm includes gradient-free [25] and gradient-based [26]. In this context, backpropagation methods are used to obtain gradients, such as automatic differentiation [27] or hybrid quantum-classical [28]. Here we use the qSGD method, which is used to avoid falling into the optimal local solution.

The task of evaluating the wave function is expressed as minimizing the *loss function* $\mathcal{L}(\omega_N^{local})$. This function is usually expressed as the expected value of the circuit, as follows

$$\mathcal{L}(\omega_N^{local}) = \langle \psi_0 | U_{l:1}^\dagger \hat{H} U_{l:1} | \psi_0 \rangle = \text{Tr}(\hat{H} U_{l:1} \rho_0 U_{l:1}^\dagger), \quad (10)$$

where $U_{l:1} = U_l(\omega_{N,l}^{local}) U_{l-1}(\omega_{N,l-1}^{local}) \cdots U_1(\omega_{N,1}^{local})$, and density operator $\rho_0 = |\psi_0\rangle\langle\psi_0|$. Each of these unitary $U_j(\omega_{N,j}^{local})$ is generated by a Pauli produce $P_j \in \{X, Y, Z\}$ as follow

$$U_j(\omega_{N,j}^{local}) = e^{-\frac{i}{2} \omega_{N,j}^{local} P_j} = \cos \frac{\omega_{N,j}^{local}}{2} I - i \sin \frac{\omega_{N,j}^{local}}{2} P_j. \quad (11)$$

Lemma 1: (Proof in Appendix A) Let unitary U_j is generated by a Pauli produce $P_j \in \{X, Y, Z\}$, that is, $U_j(\omega_{N,j}^{local}) = \exp(-i\omega_{N,j}^{local} P_j/2)$, then for arbitrary density operator ρ , we have

$$\left[-\frac{i}{2} P_j, \rho \right] = \frac{1}{2} \left(U_j \left(\frac{\pi}{2} \right) \rho U_j^\dagger \left(\frac{\pi}{2} \right) - U_j \left(-\frac{\pi}{2} \right) \rho U_j^\dagger \left(-\frac{\pi}{2} \right) \right), \quad (12)$$

where $[\cdot, \cdot]$ is the commutator.

Besides, we have theorem 1 (and the proof is shown in Appendix B).

Theorem 1: Given the density operator $\rho_0 = |\psi_0\rangle\langle\psi_0|$, and unitary $U_{l:1} = U_l(\omega_{N,l}^{local}) U_{l-1}(\omega_{N,l-1}^{local}) \cdots U_1(\omega_{N,1}^{local})$. The gradient of VQA is as follows

$$\frac{\partial}{\partial \omega_{N,j}^{local}} \mathcal{L}(\omega_N^{local}) = \frac{1}{2} \left[\mathcal{L} \left(\omega_{N,j}^{local} + \frac{\pi}{2} \right) - \mathcal{L} \left(\omega_{N,j}^{local} - \frac{\pi}{2} \right) \right]. \quad (13)$$

A similar method is used by Li *et al.* [29] and Mitarai *et al.* [30], which is called parameter shift. Just insert $\pm\pi/2$ rotation to P_j and measure the respective expected values. So let us rewrite the derivative using the chain rule as follows

$$\frac{\partial \mathcal{L}(\omega)}{\partial \omega} = \frac{\langle \hat{H} \rangle_{\omega + \frac{\pi}{2}} - \langle \hat{H} \rangle_{\omega - \frac{\pi}{2}}}{2}, \quad (14)$$

where subscript $\omega \pm \frac{\pi}{2}$ indicates parameter shift. Note that this approach evaluates the circuit twice for each parameter, which makes the analysis gradient costly. In this context, Sweke *et al.* [31] proposed the stochastic gradient descent method to compensate for the extra overhead. It is randomly selected according to the distribution of coefficient weights. Assume that $U(\omega)$ satisfies the K -term parameter shift rule. Here, we follow the approach as follow

$$\frac{\partial \langle \hat{H} \rangle_\omega}{\partial \omega_j} = \sum_{k=1}^K \gamma_{k,j} \langle \hat{H} \rangle_{\omega_{k,j}}, \quad (15)$$

where γ is the K-term coefficient, and we then have

$$\frac{\partial \mathcal{L}(\omega)}{\partial \omega_j} = \sum_{i=1}^L \sum_{k=1}^K \gamma_{k,j} \langle \hat{H}_i \rangle_{\omega_{k,j}}, \quad (16)$$

where L is the unitary operator term. The random variable is introduced as follows

$$\nabla J^{(t)}(\omega) = \sum_{i=1}^L \sum_{k=1}^K \gamma_{k,j} \hat{h}_i(\omega), \quad (17)$$

where $\hat{h}_i(\omega)$ is the n -sample mean estimator of $\langle \hat{H} \rangle_{\omega}$.

To sum up, for each iteration t , we first calculate the gradient $\nabla J^{(t)}(\omega)$. The parameter is then updated by $\omega^{(t+1)} \leftarrow \omega^{(t)} - \eta \nabla J^{(t)}(\omega)$, where η is the learning rate.

D. Server

The server then aggregates the received parameters and shares the results with all local clients. Remarkably, the server takes ω_N^{local} as input to aggregate the shared parameters and then outputs ω_N^{shared} to optimize Q_N^{local} . The form of aggregation we are considering is as follows

$$\min_{\omega \in \mathbb{R}^d} L(\omega) = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{z^i \sim \mathcal{D}^i} \mathcal{L}(\omega; z^i), \quad (18)$$

where z^i is the sample of data stored in the i -th local model. The devices of different participants are heterogeneous, and the database is non-i.i.d. So the expected values of the samples taken from different devices are as follows

$$\mathbb{E}_{z^i \sim \mathcal{D}^i} \mathcal{L}(\omega; z^i) \neq \mathbb{E}_{z^j \sim \mathcal{D}^j} \mathcal{L}(\omega; z^j). \quad (19)$$

The application of deep learning almost wholly relies on stochastic gradient descent (SGD) for optimization. Thus, it is natural to build server optimizations with SGD. We randomly select K clients in each round and calculate the loss gradient g_k under the current parameters ω_N^{local} of these clients. The server then aggregates these gradients and applies the update $\omega_N^{shared} \leftarrow \omega_N^{local} - \eta \sum_{k=1}^K \frac{1}{n} g_k$. Remarkably, each client performs qSGD using the current parameters, and the server then weighted average the results. Complete pseudo-code is given in Algorithm 2.

QFL has significant privacy advantages over a training model for centralized data. Besides, the information transmitted is the minimum update required for the model, and naturally, it does not contain local data sets with higher privacy intensity. The aggregation of model parameters does not require an update source, and updates are transmitted through a trusted third party. Our idea is that the server uses the BQC method to ensure data privacy. However, the brickwork state is large, and the BQC aggregation of the server is not fully realized in the experiment. Therefore, the method of federated average is used for the server. Further, inspired by our previous work [32], [33], model parameters can also be updated by verification of BQC or delegated quantum computation. Wahab *et al.* [34] summarized more information about aggregation techniques in Table I.

Algorithm 2: QFL.

```

1 Initialize:  $K$  is the randomly selected client per round,
    $B$  and  $E$  are the number of local batches and epochs,
   respectively.
2 Client( $k, \omega$ ):
3   Initialize:  $\hat{U}(\omega)$  and  $\hat{H}$ 
4   for epoch  $e \in E$  do
5     for batch  $b \in B$  do
6        $\omega \leftarrow \omega - \eta \nabla J(\omega; b)$ 
7     end
8   end
9   return  $\omega$  to server
10
11 Server:
12   Initialize:  $\omega_{k,1}^{shared}$ 
13   for each round  $r$  from 1 to  $R$  do
14     for each client  $k \in K$  in parallel do
15        $\omega_{k,r+1}^{local} \leftarrow \text{Client}(k, \omega_{k,r}^{shared})$ 
16     end
17      $\omega_{k,r+1}^{shared} \leftarrow \omega_{k,r}^{local} - \eta \sum_{k=1}^K \frac{1}{n} \omega_{k,r+1}^{local}$ 
18   end

```

TABLE I
COMPARE FEDERATED LEARNING AGGREGATION TECHNIQUES

Aggregation	Differences
Our	Each client performs qSGD using the current parameters, and the server then weighted average the results.
FedProx	Add a proximal term on each client device to limit the influence of each local model update on the global model.
FedPAQ	The client performs multiple local updates to the model before sharing parameters with the server.
FedMA	The permutation invariance of neurons is considered to achieve global model size adaptation.

III. EXPERIMENTS

In this section, we conduct QFL experiments on TFQ processors with some representative quantum algorithms, including VQTN, QAOA, and VQE. In the experimental environment, we used the differentiable quantum simulator TFQ [16] as the backend. The transport framework for model parameters references TensorFlow Federated [35], an open-source framework for machine learning and decentralized data computing. Specifically, we use synchronous update mode for each round of communication. At the beginning of each round, K clients are randomly selected, and each client has a private local data set. The server then sends the current global parameters to these clients. Next, each selected client performs local training based on global parameters with its local data set and sends updates to the server. Finally, the server aggregates these updates to the global parameters and repeats the process.

A. VQTN

We first train VQTN on the handwritten digital MNIST data set [24], consisting of 70,000 samples, each with 28×28 pixels images. Of these, 60,000 are used for training and 10,000 for testing. The mean image of each label from the training data set is

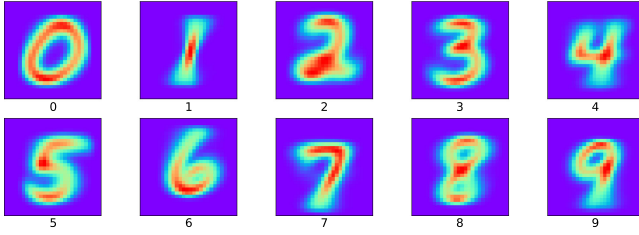


Fig. 3. The mean pixel produced by all training examples for each label. We can see that the training data contains noise. Besides, we consider that each local model satisfies non-i.i.d and unbalanced quantum data. Therefore, the digital mean pixels of each client is set differently in the experiment.

TABLE II

MEAN ACCURACY AND STANDARD DEVIATION (%) OF VQTN AND OUR ALGORITHMS ON MNIST DATA SET WITH DIFFERENT IMAGE SAMPLES

Algorithm	Clients	0 or 9	2 or 7	3 or 6
VQTN[12]	N/A	94.03 $\pm$ 0.06	96.61 $\pm$ 0.04	98.00 $\pm$ 0.05
	2	94.04 $\pm$ 0.03	96.66 $\pm$ 0.02	97.99 $\pm$ 0.08
	4	94.01 $\pm$ 0.02	96.71 $\pm$ 0.02	97.97 $\pm$ 0.04
Our	8	93.95 $\pm$ 0.09	96.72 $\pm$ 0.07	97.94 $\pm$ 0.08
	16	93.87 $\pm$ 0.14	96.56 $\pm$ 0.09	97.97 $\pm$ 0.11

shown in Fig. 3. In the NISQ era, limited by the available qubits, we reconstruct the original image as 4×4 pixels. Here, we use the area image interpolation method, and the original image is divided into different regions for interpolation calculation. In this context, we focus on non-i.i.d and unbalanced quantum data, as well as efficient communication. Here, we generate binary-class and multi-class classification tasks.

1) *Binary-Class Classification*: In this experiment, we generated three binary classification tasks. For the first task, we only keep images containing 0 or 9 (i.e., 2 or 7, and 3 or 6 are the second and last tasks, respectively). Training data is assigned to 2, 4, and 8 local models for each task, which satisfy non-i.i.d and unbalanced quantum data. In each training round, the batch size is $B = 20$, and the epoch is $E = 1$. To highlight the performance of decentralized data, we added a comparative experiment of the centralized data set and compared our approach with the state-of-the-art VQTN method. Remarkably, we performed 20 training rounds, and the mean accuracy and standard deviation are calculated from the results of 5 experiments.

In this work, the mean and standard deviation of the classification accuracy of the two algorithms on the MNIST data set are given in Table II. The experimental data is based on the results of five experiments, and the bold values indicate the best results for each task. The ‘Algorithm’ column describes VQTN and our proposed algorithm. The ‘Clients’ column describes the number of clients for collaborative computing. Firstly, we notice from three rows of the experimental results that our method improves the average accuracy of tasks ‘0 or 9’ and ‘2 or 7’. Secondly, the 2 client has similar classification accuracy to the traditional method in task 1. Thirdly, the average classification accuracy of the 8 clients showed an advantage in task 2. Finally, the last task reveals that the average accuracy of VQTN is superior to the multi-client approach. Thus, Table II indicates that our method achieves excellent performance on the MNIST data set, comparable to the level of non-federated training.

TABLE III
THE EVALUATION METRICS OF THE TWO ALGORITHMS

Methods	Clients	0 or 9	2 or 7	3 or 6
	N/A	0.9820	0.9905	0.9972
	2	0.9816	0.9905	0.9972
	4	0.9820	0.9906	0.9975
	8	0.9822	0.9906	0.9969
	16	0.9813	0.9905	0.9969
	N/A	0.9399	0.9664	0.9806
	2	0.9400	0.9669	0.9804
	4	0.9403	0.9675	0.9803
	8	0.9391	0.9674	0.9800
	16	0.9386	0.9659	0.9803

TABLE IV
THE EVALUATION METRICS OF MULTI-CLASS CLASSIFICATION

Methods	2	3	6	7
Precision	0.7343	0.8298	0.8564	0.8484
Recall	0.7045	0.7673	0.8841	0.9202
F1-score	0.7191	0.7973	0.8701	0.8829
Accuracy	0.8180			

We also observe the evaluation metrics of the above algorithm, which further reflects the performance of our proposed method. Here, the area under the curve (AUC) and F1-score are used to evaluate the quality of classification, which are the most widely used evaluation metrics. Intuitively, the higher the metric, the better the performance. In this work, detailed results of the two algorithms are presented in Table III. The experimental data is based on the results of five experiments, and the bold values indicate the best results for each task.

Table III indicates that our approach has achieved excellent performance in two evaluation metrics of three tasks and is superior to non-federated methods on the MNIST data set. Firstly, we noticed from the AUC of the experimental results, and our approach has a high evaluation value in clients 4 and 8. It indicates that the classification effect of QFL is superior to non-federated training. Secondly, the F1-score of QFL is better than non-federated, indicating excellent precision of the QFL model in tasks 1 and 2. To sum up, the performance of the QFL algorithm exhibits advantages in decentralized data training.

2) *Multi-Class Classification*: In this experiment, we extend QFL to multi-class classification tasks. We only keep images containing ‘2, 3, 6 or 7’ for the training task, and the data is assigned to the 4 local models. We performed 20 training rounds. In each round, the batch size is $B = 20$, and the epoch is $E = 1$. In this work, we analyzed the evaluation metrics of multi-class classification, and the detailed results of the QFL algorithm are presented in Table IV. The bold values indicate the best results for each task.

The ‘Precision’ describes the number of correct samples divided by the number of all samples. The ‘Recall’ describes the true positive rate in the predicted sample. The ‘Accuracy’ describes the mean of the classification accuracy. For this task, the higher the metric, the better the performance. From the experimental results in Table IV, we observed that the mean accuracy of 4-class tasks was 81.80%. Besides, recall and F1-score indicators show that label 7 performed better than the others. In contrast, label 6 has the best precision. Table IV indicates

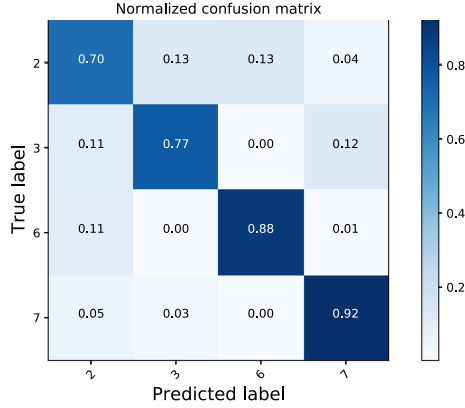


Fig. 4. Illustration of the ‘2, 3, 6 or 7’ classification confusion matrix of QFL algorithm in the MNIST data set.

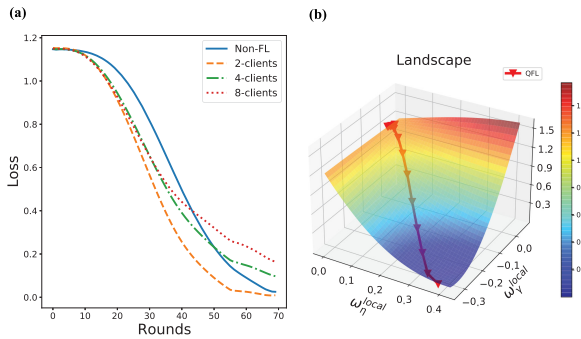


Fig. 5. Illustration of the loss and gradient descent of QFL and Non-FL in random graph data sets. (a) Target loss of parameter evolution in training rounds. For each training round, solid, dashed, dash-dot, and dotted lines represent Non-FL, 2-clients, 4-clients, and 8-clients, respectively. (b) The random initial point of QFL flows to the minimum value in the parametric landscape.

that our approach has achieved excellent performance in these evaluation metrics of 4-class tasks. In order to visualize the details of testing performance, the normalized confusion matrix is shown in Fig. 4.

Fig. 4 indicates that the numbers 6 and 7 are easy to distinguish. It is consistent with the indicators in Table IV. However, the number of 2 may be incorrectly recognized as 3 or 6. It is limited by the expressive power of the shallow quantum circuit model. To sum up, the proposed QFL algorithm can carry out multi-classification and achieve a similar performance as the non-federated algorithm.

B. QAOA

We then turn to consider the QAOA model, which evaluates the wave function to solve the bit string. A Simple example of QFL solving a 3-regular graph is given in Appendix C. Here, we randomly generate 100 samples of the six qubits random graph $\mathcal{G}$ as the training data set. For all experiments, we performed 70 rounds. To avoid barren plateaus [36], we initialize the parameter 0.01 and the learning rate $\eta = 0.01$. In each round, the batch size is $B = 16$, and the epoch is $E = 1$. Thus, the exponentials in the form of (4) are $P = 1$. The training loss and gradient descent of parameter landscape are shown in Fig. 5. Remarkably, as the number of rounds increases, the final loss converges to a scalar.

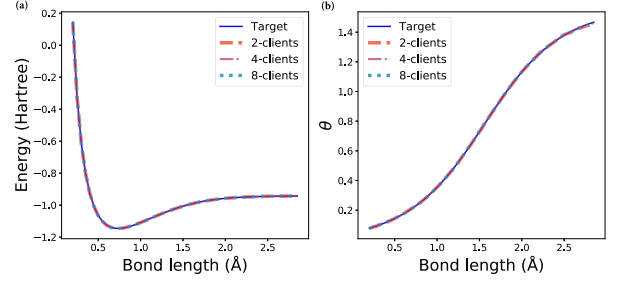


Fig. 6. Illustration of target energies and variational parameters of QFL in molecular hydrogen. In both figures, target, 2-clients, 4-clients, and 8-clients are indicated by solid, dashed, dash-dot, and dotted lines, respectively. (a) Target energies and optimization results for the QFL. (b) The corresponding variational parameter of each round.

In the first 42 rounds, QFL losses declined faster than the non-federated algorithm, indicating the advantage of distributed data training, as shown in Fig. 5(a). However, in the remaining finite rounds, the multi-client convergence is worse than non-federated. Nevertheless, QFL still shows an excellent gradient descent in Fig. 5(b). The results show that QFL can find the optimal path in six qubits random graph. In such a training process, QFL can cooperate with multiple clients to solve the MaxCut problem.

C. VQE

We now turn to consider the VQE model, which computes wave functions to solve quantum chemistry problems. Here, we use QFL to calculate the target energy and variational parameters of the molecular hydrogen. We consider the Hamiltonian involving two qubits, following the scheme in Ref. [37]. Using an STO-3G basis, we have

$$H_{H_2} = c_0 I + c_1 \sigma_0^z + c_2 \sigma_1^z + c_3 \sigma_0^z \sigma_1^z + c_4 \sigma_0^x \sigma_1^x + c_5 \sigma_0^y \sigma_1^y, \quad (20)$$

where c_i is the coefficients depend on the bond length. The unitary coupled cluster (UCC) ansatz is $U(\theta) = \exp(-i\theta \sigma_0^x \sigma_1^y)$. Besides, the Hartree-Fock state is $|01\rangle$.

In this work, 54 bond lengths were uniformly chosen from 0.25 to 2.85 a.u. to calculate Hamiltonians as training samples. We extract all samples from the original data and assign them to 2, 4, and 8 local models, satisfying non-i.i.d and unbalanced quantum data. In each training round, the batch size is $B = 16$, and the epoch is $E = 1$. In order to highlight the performance of QFL, the ground state energies and variational parameters are shown in Fig. 6. Remarkably, we performed 180 training rounds, the target energies and variational parameters are calculated at different bond lengths.

The results show that QFL can calculate the ground state energy cooperatively, as shown in Fig. 6(a). At the same time, the variational parameter evolves to the optimal target value, as shown in Fig. 6(b). In such a process, QFL can cooperate with multiple clients to modify the model parameters (or hyperparameters) to solve quantum chemistry problems.

IV. DISCUSSION AND CONCLUSIONS

Here we propose communication-efficient learning of VQA from decentralized data. As the first step in federated learning

to protect VQA privacy, it is expected that QFL can be tested on other general problems. In quantum neural networks, we use a state-of-the-art VQTN model for classification experiments. It requires only 20 rounds of communication to achieve non-federated similar performance. In quantum chemistry, we follow UCC ansatz and consider molecular hydrogen with a single variational parameter. However, for some complex quantum chemistry problems, we can further study in the future. In quantum approximation optimization, we solve the MaxCut problem of 3-regular and random graphs. In such a training process, QFL can cooperate with multiple clients to solve the MaxCut problem. It is expected that QFL will help us improve the privacy security of VQA. In the future, we will study the QFL of transfer learning, which inspires investigations in the field of secure QML.

The difference between quantum data and classical data in model training is that they are applicable to different application scenarios of local models. There may also be privacy issues in the use of quantum data. In this case, BQC can be introduced to the server to avoid third-party theft. Both Classic FL and Quantum FL are designed to protect user data privacy. The difference is that FL uses classical data communication to deal with the classical local model. QFL uses BQC communication to process quantum models.

We demonstrate that the proposed QFL algorithm has an excellent performance in the variational task of decentralized data. Furthermore, the QFL has been used to solve some representative quantum algorithms, including VQTN, QAOA, and VQE. Remarkably, privacy and security risks are significantly reduced by limiting possible attacks on clients with scattered data.

APPENDIX A PROOF OF LEMMA 1

Proof: Firstly, following property of the commutator, we have

$$\begin{aligned}
 \left[-\frac{i}{2}P_j, \rho \right] &= \left(-\frac{i}{2}P_j \cdot \rho \right) - \rho \cdot \left(-\frac{i}{2}P_j \right) \\
 &= \frac{1}{2} [(-iP_j)\rho - \rho(-iP_j)] \\
 &= \frac{1}{4} [(I\rho I + I\rho(iP_j) + (-iP_j)\rho I \\
 &\quad + (-iP_j)\rho(iP_j)) - h.c.] \\
 &= \frac{1}{4} [(I - iP_j)\rho(I + iP_j) - h.c.]. \quad (21)
 \end{aligned}$$

Then based on (11), we have

$$U_j \left(\frac{\pi}{2} \right) = \frac{1}{\sqrt{2}} (I - iP_j). \quad (22)$$

On the basis of (22), we reformulated the original equation as follows

$$\left[-\frac{i}{2}P_j, \rho \right] = \frac{1}{2} \left(U_j \left(\frac{\pi}{2} \right) \rho U_j^\dagger \left(\frac{\pi}{2} \right) - U_j \left(-\frac{\pi}{2} \right) \rho U_j^\dagger \left(-\frac{\pi}{2} \right) \right). \quad (23)$$

□

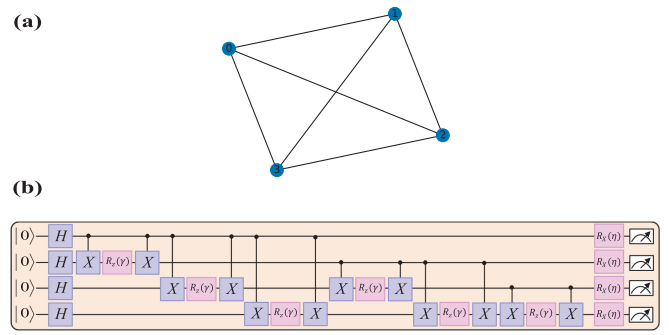


Fig. 7. Illustration of the Maxcut problem. (a) Random 3-regular graph with 4 qubits. (b) The corresponding quantum circuit of G .

APPENDIX B PROOF OF THEOREM 1

Proof: According to (10), the gradient is calculated as follows

$$\begin{aligned}
 \frac{\partial}{\partial \omega_{N,j}^{local}} \mathcal{L}(\omega_N^{local}) &= \text{Tr} \left(\hat{H} U_{l:j+1} \frac{\partial U_j}{\partial \omega_{N,j}^{local}} U_{j-1:1} \rho_0 U_{l:1}^\dagger \right. \\
 &\quad \left. + \hat{H} U_{l:1} \rho_0 U_{j-1:1}^\dagger \frac{\partial U_j^\dagger}{\partial \omega_{N,j}^{local}} U_{l:j+1}^\dagger \right) \\
 &= \text{Tr} \left(\hat{H} U_{l:j+1} \left(-\frac{i}{2} P_j \right) \rho_j U_{l:j+1}^\dagger \right. \\
 &\quad \left. - \hat{H} U_{l:j+1} \rho_j \left(-\frac{i}{2} P_j \right) U_{l:j+1}^\dagger \right) \\
 &= \text{Tr} \left(\hat{H} U_{l:j+1} \left[-\frac{i}{2} P_j, \rho_j \right] U_{l:j+1}^\dagger \right), \quad (24)
 \end{aligned}$$

where $\rho_j = U_{j:1} \rho_0 U_{j:1}^\dagger$. Based on Lemma 1, we reformulated the original equation as follows

$$\begin{aligned}
 \frac{\partial}{\partial \omega_{N,j}^{local}} \mathcal{L}(\omega_N^{local}) &= \frac{1}{2} \text{Tr} \left[\hat{H} U_{l:j+1} U_j \left(\frac{\pi}{2} \right) \rho_j U_j^\dagger \left(\frac{\pi}{2} \right) U_{l:j+1}^\dagger \right] \\
 &\quad - \frac{1}{2} \text{Tr} \left[\hat{H} U_{l:j+1} U_j \left(-\frac{\pi}{2} \right) \rho_j U_j^\dagger \left(-\frac{\pi}{2} \right) U_{l:j+1}^\dagger \right] \\
 &= \frac{1}{2} \left[\mathcal{L} \left(\omega_{N,j}^{local} + \frac{\pi}{2} \right) - \mathcal{L} \left(\omega_{N,j}^{local} - \frac{\pi}{2} \right) \right]. \quad (25)
 \end{aligned}$$

□

APPENDIX C QFL FOR MAXCUT PROBLEM

In this appendix, we demonstrate the QAOA task of 4 qubits, where QFL is used to solve the Maxcut problem. For the training task, we randomly generate a 3-regular graph G as the training data set, and the data is assigned to the 2 local models. Graph G and the corresponding quantum circuit are shown in Fig. 7. The exponentials in the form of (4) is $P = 1$.

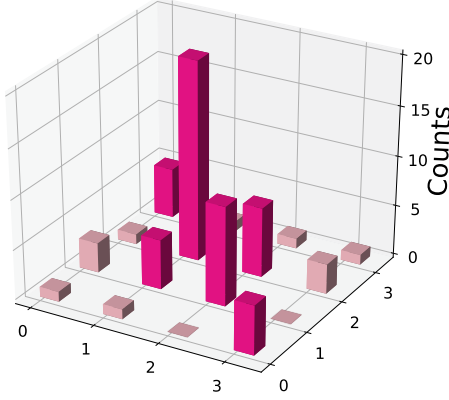


Fig. 8. Illustration of the sample results. There are six final quantum states of the solution satisfying $\mathcal{G}$, with counts of 5, 5, 10, 20, 7, and 5.

As Definition 2, we compute the objective function as follows

$$\begin{aligned}
 Q_N^{local} &= e^{-i\eta\hat{H}_M} e^{-i\gamma\hat{H}_C} |\psi_0\rangle^{\otimes n} \\
 &= e^{-i\eta\sum_{j\in V}\sigma_j^x} e^{-i\gamma\sum_{\{j,k\}\in E}\frac{1}{2}(I-\sigma_j^z\sigma_k^z)} \\
 &= \prod_j e^{-i\eta\sigma_j^x} \cdot \prod_{j,k} e^{-i\frac{\gamma}{2}\sigma_j^I\otimes\sigma_k^I} \cdot \prod_{j,k} e^{i\frac{\gamma}{2}\sigma_j^z\otimes\sigma_k^z} \\
 &= R_x(j, 2\eta) \cdot I(\{j, k\}, \gamma) \cdot \prod_{j,k} e^{i\frac{\gamma}{2}\sigma_j^z\otimes\sigma_k^z}, \quad (26)
 \end{aligned}$$

where the first two terms are the X -rotation gate and constant terms for γ , respectively. In addition, the third term can be written as follows

$$\begin{aligned}
 &\prod_{j,k} e^{i\frac{\gamma}{2}\sigma_j^z\otimes\sigma_k^z} \\
 &= \begin{bmatrix} e^{i\frac{\gamma}{2}} & 0 & 0 & 0 \\ 0 & e^{-i\frac{\gamma}{2}} & 0 & 0 \\ 0 & 0 & e^{-i\frac{\gamma}{2}} & 0 \\ 0 & 0 & 0 & e^{i\frac{\gamma}{2}} \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} e^{i\frac{\gamma}{2}} & 0 & 0 & 0 \\ 0 & e^{-i\frac{\gamma}{2}} & 0 & 0 \\ 0 & 0 & e^{i\frac{\gamma}{2}} & 0 \\ 0 & 0 & 0 & e^{-i\frac{\gamma}{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
 &= CNOT(j, k) \{I_j \otimes R_z(k, -\gamma)\} CNOT(j, k). \quad (27)
 \end{aligned}$$

In combination with Eqs. (26) and (27), we know that the quantum circuit of Q_N^{local} consists of $CNOT$, R_z , and R_x . Similarly, a quantum circuit of a 3-regular graph $\mathcal{G}$ is shown in Fig. 7(b).

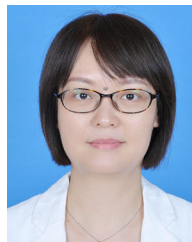
In this work, we performed 50 training rounds. Besides, the batch size is $B = 1$, and the epoch is $E = 1$. Sampling results repeated 64 times are shown in Fig. 8. The final quantum states satisfying the MaxCut problem are $|0011\rangle$, $|0101\rangle$, $|0110\rangle$, $|1001\rangle$, $|1010\rangle$, and $|1100\rangle$. Fig. 8 indicates that the sum of the correct solutions is 81.25%. The results show that we obtain

the solution of the MaxCut problem with high probability. Remarkably, QFL can collaborate with multiple clients to optimize quantum circuits.

REFERENCES

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist.* A. Singh and J. Zhu, Eds., Fort Lauderdale, FL, USA, PMLR, 2017, vol. 54, pp. 1273–1282.
- [2] K. Bonawitz *et al.*, "Practical secure aggregation for privacy-preserving machine learning," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, New York, NY, USA, 2017, pp. 1175–1191.
- [3] S. Hardy *et al.*, "Private federated learning on vertically partitioned data via entity resolution and additively homomorphic encryption," 2017. [Online]. Available: <https://arxiv.org/abs/1711.10677>
- [4] Y. Liu, Y. Kang, C. Xing, T. Chen, and Q. Yang, "A secure federated transfer learning framework," *IEEE Intell. Syst.*, vol. 35, no. 4, pp. 70–82, Jul./Aug. 2020.
- [5] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, pp. 1–19, Jan. 2019.
- [6] P. Wittek, *Quantum Machine Learning: What Quantum Computing Means to Data Mining*. Burlington, MA, USA: Elsevier Science, 2014.
- [7] S. Lloyd, M. Mohseni, and P. Rebentrost, "Quantum principal component analysis," *Nature Phys.*, vol. 10, pp. 631–633, Sep. 2014.
- [8] R. Huang, X. Tan, and Q. Xu, "Quantum algorithm for hyperparameters estimation," *Quantum Sci. Technol.*, vol. 5, Aug. 2020, Art. no. 045008.
- [9] V. Giovannetti, S. Lloyd, and L. Maccone, "Quantum random access memory," *Phys. Rev. Lett.*, vol. 100, Apr. 2008, Art. no. 160501.
- [10] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, Aug. 2018, Art. no. 79.
- [11] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, "Parameterized quantum circuits as machine learning models," *Quantum Sci. Technol.*, vol. 4, Nov. 2019, Art. no. 043001.
- [12] R. Huang, X. Tan, and Q. Xu, "Variational quantum tensor networks classifiers," *Neurocomputing*, vol. 452, pp. 89–98, 2021.
- [13] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," 2014. [Online]. Available: <https://arxiv.org/abs/1411.4028>
- [14] A. Peruzzo *et al.*, "A variational eigenvalue solver on a photonic quantum processor," *Nature Commun.*, vol. 5, Jul. 2014, Art. no. 4213.
- [15] D.-B. Zhang and T. Yin, "Collective optimization for variational quantum eigensolvers," *Phys. Rev. A*, vol. 101, Mar. 2020, Art. no. 032311.
- [16] M. Broughton *et al.*, "Tensorflow quantum: A software framework for quantum machine learning," 2020. [Online]. Available: <https://arxiv.org/abs/2003.02989>
- [17] W. Li, S. Lu, and D.-L. Deng, "Quantum federated learning through blind quantum computing," *Sci. China Phys., Mechanics Astron.*, vol. 64, Sep. 2021, Art. no. 100312.
- [18] S. Y.-C. Chen and S. Yoo, "Federated quantum machine learning," *Entropy*, vol. 23, no. 4, 2021, Art. no. 460.
- [19] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014. [Online]. Available: <https://arxiv.org/abs/1412.6980>
- [20] E. Grant *et al.*, "Hierarchical quantum classifiers," *NPJ Quantum Inf.*, vol. 4, Dec. 2018, Art. no. 65.
- [21] W. Huggins, P. Patil, B. Mitchell, K. B. Whaley, and E. M. Stoudenmire, "Towards quantum machine learning with tensor networks," *Quantum Sci. Technol.*, vol. 4, Jan. 2019, Art. no. 024001.
- [22] D. Liu *et al.*, "Machine learning by unitary tensor network of hierarchical tree structure," *New J. Phys.*, vol. 21, Jul. 2019, Art. no. 073059.
- [23] S. Caldas *et al.*, "LEAF: A benchmark for federated settings," 2018. [Online]. Available: <https://arxiv.org/abs/1812.01097>
- [24] Y. LeCun, C. Cortes, and C. J. Burges, "The MNIST database of handwritten digits," 1998. [Online]. Available: <http://yann.lecun.com/exdb/mnist/>
- [25] C. Kokail *et al.*, "Self-verifying variational quantum simulation of lattice models," *Nature*, vol. 569, pp. 355–360, May 2019.
- [26] J.-G. Liu, Y.-H. Zhang, Y. Wan, and L. Wang, "Variational quantum eigensolver with fewer qubits," *Phys. Rev. Res.*, vol. 1, Sep. 2019, Art. no. 023025.
- [27] A. G. Baydin, B. A. Pearlmutter, A. A. Radul, and J. M. Siskind, "Automatic differentiation in machine learning: A survey," *J. Mach. Learn. Res.*, vol. 18, pp. 5595–5637, Jan. 2017.

- [28] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran, "Evaluating analytic gradients on quantum hardware," *Phys. Rev. A*, vol. 99, Mar. 2019, Art. no. 032331.
- [29] J. Li, X. Yang, X. Peng, and C.-P. Sun, "Hybrid quantum-classical approach to quantum optimal control," *Phys. Rev. Lett.*, vol. 118, Apr. 2017, Art. no. 150503.
- [30] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, "Quantum circuit learning," *Phys. Rev. A*, vol. 98, Sep. 2018, Art. no. 032309.
- [31] R. Sweke *et al.*, "Stochastic gradient descent for hybrid quantum-classical optimization," *Quantum*, vol. 4, Aug. 2020, Art. no. 314.
- [32] Q. Xu, X. Tan, and R. Huang, "Improved resource state for verifiable blind quantum computation," *Entropy*, vol. 22, no. 9, 2020, Art. no. 996.
- [33] Q. Xu, X. Tan, R. Huang, and M. Li, "Verification of blind quantum computation with entanglement witnesses," *Phys. Rev. A*, vol. 104, Oct. 2021, Art. no. 042412.
- [34] O. A. Wahab, A. Mourad, H. Otrouk, and T. Taleb, "Federated machine learning: Survey, multi-level classification, desirable criteria and future directions in communication and networking systems," *IEEE Commun. Surv. Tut.*, vol. 23, no. 2, pp. 1342–1397, Apr.–Jun. 2021.
- [35] The TensorFlow Federated Authors, "TensorFlow Federated," 2018. [Online]. Available: <https://github.com/tensorflow/federated>
- [36] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nature Commun.*, vol. 9, Nov. 2018, Art. no. 4812.
- [37] C. Hempel *et al.*, "Quantum chemistry calculations on a trapped-ion quantum simulator," *Phys. Rev. X*, vol. 8, Jul. 2018, Art. no. 031022.



Xiaoqing Tan received the Ph.D. degree from Sun Yat-sen University, Guangzhou, China, in 2004. She is currently a Professor with the College of Information Science and Technology, College of Cyber Security, Jinan University, Guangzhou. Her research interests include secure quantum computing and quantum machine learning.



Qingshan Xu received the B.E. degree from China Jiliang University, Hangzhou, China, in 2018. He is currently working toward the M.Sc. degree with Jinan University, Guangzhou, China. His research interests include secure quantum computing.



Rui Huang received the B.E. degree from Huizhou University, Huizhou, China, in 2015 and the M.E. degree from the Guilin University of Technology, Guilin, China, in 2017. He is currently working toward the Doctoral degree with Jinan University, Guangzhou, China. His research interests include secure quantum computing and quantum machine learning.